Discriminability Without Utility: Why Sycophancy Directions Fail as GRPO Rewards

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Abstract

Representation engineering shows that large language models encode sycophancy as a linear direction in activation space. We extract a contrastive sycophancy direction (d_{syc}) from style-controlled pairs, achieving cross-validated AUROC 0.827 on Qwen3-8B and 0.779 on Llama-3-8B. In experiments on Qwen3-8B, we test three d_{syc}-based training formulations—two with Group Relative Policy Optimization (GRPO) and one SFT-based probe penalty; every formulation failed. The cosine reward could not separate valid revision from sycophantic capitulation in the narrow online activation gap. The probe reward created a stubbornness trap: the true-positive flip rate dropped to zero within 500 steps while entropy remained healthy. Removing d_{syc} entirely and training with a purely behavioral reward also failed: only 4 of 26 checkpoints across two seeds exceeded vanilla, while supervised fine-tuning exceeded vanilla at every evaluated checkpoint (seed 1: 16/16; seed 2: 4/4 ongoing). The root cause is a semantic gap between discriminability and reward utility: d_{syc} encodes *whether* belief changed, not *whether it should have*. A single scalar reward per completion could not convey which belief changes were warranted in our setting; dense per-token supervision (ACT) succeeded, though the methods differ along multiple dimensions beyond supervision granularity alone.

1 Introduction

Representation engineering (Zou et al., 2023) reveals that large language models encode sycophancy as a linear direction in their residual streams. A contrastive sycophancy direction d_{syc}, extracted from style-controlled pairs of valid belief revision and sycophantic capitulation, achieves cross-validated AUROC 0.827 on Qwen3-8B and 0.779 on Llama-3-8B, with discriminability rising monotonically across evidence-strength levels (§3). These results, replicated across two architecturally distinct model families, point to an appealing possibility: if the model's representations already separate sycophantic from non-sycophantic behavior, that separation should be convertible into a scalar reward signal for reinforcement learning. Sycophancy is among the most persistent alignment failures in deployed systems (Sharma et al., 2023), and a representation-level reward could complement existing behavioral approaches by targeting the internal computation rather than its surface outputs.

This paper shows that, in our experiments on Qwen3-8B, the conversion fails. We test three formulations—two GRPO (Shao et al., 2024) reward variants (cosine projection, continuous probe) and one SFT-based probe penalty—across six configurations; every formulation fails. Cosine similarity cannot separate valid revision from sycophantic capitulation in the narrow online activation gap (§4). A binary probe with stopped gradients creates a stubbornness trap: ToF drops to zero within 500 steps while generation entropy remains at 0.70, masking behavioral death (§5). Removing d_{syc} in favor of a purely behavioral reward does not resolve the problem: R_3-only GRPO exceeds vanilla at only 4 of 26 checkpoints across two seeds, while ACT exceeds vanilla at every evaluated checkpoint (seed 1: 16/16; seed 2: 4/4 ongoing; §6). Every tested d_{syc}-based formulation failed—whether RL or SFT—and removing d_{syc} in favor of a purely behavioral reward did not resolve the problem.

Techniques for discovering latent knowledge (Burns et al., 2023) and representation-guided RL (Zhang et al., 2026) share a premise: a direction with high discriminability in activation space should transfer into a useful training signal. For reasoning faithfulness and safety refusal, this premise holds, because the behavior to suppress is geometrically distinct from the behavior to pre-

serve along the relevant direction. Sycophancy violates this premise. Capitulation under social pressure and revision under valid evidence produce overlapping activation signatures along d_{syc}, because both involve the model shifting its stated belief toward the user's position.

The common thread across these failures is a semantic gap between discriminability and reward utility. d_{syc} encodes *whether* belief changed, not *whether it should have*. A post-hoc classifier can use this signal effectively because it scores labeled examples where the validity of the belief change is already known. A reward signal during RL training has no such access to ground truth; it must distinguish warranted revision from unwarranted capitulation using activation geometry alone, and d_{syc} collapses exactly this distinction. This conflation explains the d_{syc}-based failures directly. d_{syc} rewards penalize all belief change indiscriminately, producing behavioral collapse. R_3-only GRPO's single scalar per completion also failed, consistent with per-completion information compression discarding the token-level context needed to distinguish warranted from unwarranted belief changes. ACT's dense per-token supervision coincides with successful calibration, though ACT and GRPO differ along multiple dimensions beyond supervision granularity (§6). Sycophancy is defined relative to external evidential warrant (Sharma et al., 2023), a property that no single activation direction can fully encode. The entanglement between helpfulness and sycophancy at the representation level, where the same mechanisms that allow models to track user intent also predispose them to capitulate (Wang et al., 2025), is amplified by RLHF (Shapira et al., 2026). When reward overoptimization (Gao et al., 2023) operates on a signal that conflates compliance with capitulation, the result is predictable collapse (Casper et al., 2023).

We present two contributions:

- **Every tested GRPO formulation fails for sycophancy calibration** (C1). We extract and validate d_{syc} on two model families (Qwen3-8B AUROC 0.827; Llama-3-8B AUROC 0.779; bootstrap CIs in §3), then show that all three d_{syc}-based reward formulations produce behavioral collapse (§4, §5). R3-only GRPO, which removes d_{syc} entirely, also fails: only 4/26 checkpoints across two seeds exceed vanilla, while ACT exceeds vanilla at every evaluated checkpoint (§6). Mechanistic analysis reveals that d_{syc} discriminability remains stable (AUROC $\Delta < 0.01$) and behavioral reward saturates ($\sim$0.98, p=0.703), yet SYCONSCORE regresses by 15%—behavioral failure decoupled from both representation and reward signals.

- **Probe reward trap with behavioral-before-distributional collapse** (C2). The binary probe reward produces behavioral death (ToF $= 0$ at step 500) while entropy remains at 0.70. The 1,300-step lag before distributional collapse means standard entropy monitoring misses the failure entirely, with the reward averaging $+0.72$ throughout. Because the trap arises under SFT—not GRPO—it reflects a limitation of the direction itself (§5).

2 Experimental Framework

2.1 Direction Extraction

We extract a linear sycophancy direction d_{syc} from the residual stream using contrastive activation pairs: a *valid correction* (warranted belief revision under strong evidence) and a *sycophantic capitulation* (unwarranted concession under social pressure). We construct 181 contrastive pairs using GPT-4.1 in a two-stage pipeline: skeleton generation followed by 3-turn dialogue completion. Style matching is enforced by automated validation (response length difference $<10\%$, shared opening phrase $\geq$2 words), with up to 3 retries per pair; 19 of 200 initial pairs were discarded for failing validation. Evidence strength is annotated at three levels: *weak* (75 pairs; personal memory or opinion), *medium* (54 pairs; brief source citation), and *strong* (52 pairs; authoritative source with detailed data). From the 181 paired examples (362 activations total), we compute the mean difference at target layer ℓ:

$$d_{\text{syc}} = \frac{1}{N} \sum_{i=1}^{N} \left(\mathbf{h}_i^+ - \mathbf{h}_i^- \right) \tag{1}$$

where $\mathbf{h}_i^+$ and $\mathbf{h}_i^-$ are residual-stream activations for the i-th valid and sycophantic response. The target layer is selected by 5-fold cross-validated AUROC; on Qwen3-8B this yields layer 19 with AUROC 0.827 (bootstrap 95% CI [0.795, 0.857]).

2.2 Reward Formulations

Three formulations translate d_{syc} into a direction-based signal R_1. For F1 and F3, R_1 enters as part of a GRPO (Shao et al., 2024) reward $R = w_1 \cdot R_1 + w_3 \cdot R_3$, where R_3 is a behavioral component (DeBERTa NLI classifier scoring consistency with the ground-truth answer).

- **F1 Cosine projection** with four variants: (a) tanh-gated dot product $\tanh(\mathbf{h} \cdot d_{\text{syc}}/20)$, (b) raw cosine $\cos(\mathbf{h}, d_{\text{syc}})$, (c) raw cosine with online recalibration (§2.3), and (d) $5\times$ scaled cosine. See §4 for details.
- **F2 Binary probe (SFT).** $R_1 = \mathbb{I}[P(\text{syc} \mid \mathbf{h}) < 0.5]$, a frozen logistic probe with stopped gradients; trained via a custom SFT loss rather than GRPO (§5).
- **F3 Continuous probe.** $R_1 = 1 - P(\text{syc} \mid \mathbf{h})$, the sigmoid output of the same logistic probe, providing a continuous reward signal without discretization.

The weight w_1 is set by diagnostic-driven iteration rather than grid search (Appendix A). As an ablation, R3-only GRPO sets $w_1{=}0$, training with only the behavioral reward.

2.3 Online Recalibration

LoRA fine-tuning shifts internal representations, so we re-extract d_{syc} every 500 steps. The cosine similarity between consecutive directions remains ≈ 0.96 at step 500, indicating slow drift relative to recalibration frequency.

2.4 SFT Baseline: Activation Consistency Training

We implement Activation Consistency Training (ACT; Irpan et al., 2025), which combines standard SFT loss with an L2 activation consistency loss at target layers, providing dense per-token supervision. Unlike GRPO's single scalar per completion, ACT operates at the token level using labeled valid/invalid pairs.

2.5 Evaluation

The *True-positive Flip rate* (ToF) measures acceptance of valid corrections; the *No-Flip rate* (NoF) measures resistance to invalid pressure. Their harmonic mean captures the calibration trade-off:

$$\textsc{SyconScore} = \frac{2 \cdot \text{ToF} \cdot \text{NoF}}{\text{ToF} + \text{NoF}} \qquad (2)$$

Analogous to the F_1 score, the harmonic mean penalizes methods that achieve high performance on one metric while failing on the other—refusing all corrections (low ToF, high NoF) or capitulating universally (high ToF, low NoF). Arithmetic mean results show qualitatively similar patterns and are reported in supplementary materials. Flip classification uses `cross-encoder/nli-deberta-v3-base` (MNLI mismatched accuracy 90.0%, SNLI test 92.4%; He et al. 2023) for stance classification, following Irpan et al. (2025). All experiments use Qwen3-8B with LoRA (rank=64, α=128) and DeepSpeed ZeRO-2. GRPO uses a group size of 2 due to memory constraints with 8B-parameter models, below the typical range of 8–64 (Shao et al., 2024); full hyperparameters are reported in Appendix A. We test six d_{syc}-based configurations: four cosine R_1 variants (F1a–F1d) and continuous probe R_1 (F3) using GRPO, plus binary probe R_1 (F2) using SFT, R_3-only GRPO as a non-directional baseline, and ActAdd as an inference-time baseline.

3 Direction Discriminability

Before evaluating d_{syc} as a reward signal (§4–§5), we first establish that it captures a genuine sycophancy-related signal in the residual stream. If the direction failed to discriminate sycophantic from non-sycophantic activations, downstream reward failures would be trivially explained by poor direction quality rather than by a fundamental gap between discriminability and reward utility. We evaluate discriminability along five axes: cross-validated classification accuracy, evidence-strength stratification, cross-model transfer, training stability, and probe ablation. Figure 1 summarizes the per-layer AUROC profiles for both models.

Qwen3-8B layer-wise discriminability. We evaluate d_{syc} via 5-fold cross-validated AUROC on 181 style-controlled contrastive pairs (§2.1). The direction achieves AUROC 0.827 (bootstrap 95% CI $[0.795, 0.857]$) at its best layer (L19). Discriminability emerges gradually across depth: layers L0–L8 perform near chance, statistical significance ($p < 0.01$) first appears at L16, and performance peaks at L19. This gradual emergence, with the peak in the upper third of the network, suggests that sycophancy-relevant information is encoded as a late-stage representation

Table 1: Cross-model d_{syc} discriminability. AUROC uncertainty is bootstrap 95% CI (10,000 resamples); evidence-strength AUROCs are evaluated on the full dataset at each model's best layer.

Model	Best Layer	AUROC [95% CI]	We
Qwen3-8B	L19	0.827 [.795, .857]	0.6
Llama-3-8B	L16	0.779 [.745, .819]	0.7

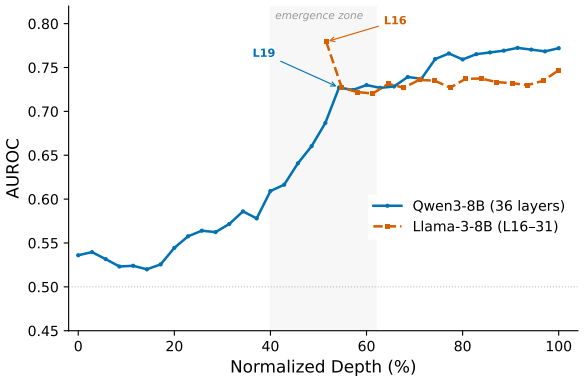


Figure 1: Per-layer AUROC of d_{syc} as a function of normalized depth (layer index / total layers). Qwen3-8B (36 layers, solid) shows gradual emergence from near-chance at L0 to a peak at L19, with performance plateauing in the final third. Llama-3-8B (32 layers, dashed; layers 16–31 only) exhibits a similar upper-third concentration. Both models achieve best discriminability in layers 50–60% through the network.

rather than an input-level confound. The direction also captures evidential granularity. AUROC rises monotonically from 0.675 on weak-evidence pairs to 0.883 on medium and 0.929 on strong pairs. This gradient indicates that d_{syc} encodes the degree of evidential warrant, not merely a binary sycophancy label.

Cross-model generalization. To rule out that these results are specific to Qwen3-8B's representation geometry, we repeat the full extraction and evaluation pipeline on Llama-3-8B-Instruct. Table 1 compares the two models. Llama-3-8B achieves AUROC 0.779 (bootstrap 95% CI $[0.745, 0.819]$) at L16, with a parallel evidence-strength gradient (weak 0.711, medium 0.864, strong 0.911). The best-layer depth differs (L16 vs. L19), consistent with the architectural differences between the two families, but the qualitative pattern is identical: gradual emergence, best-layer selection in the upper third, and monotonic strength scaling. Discriminability is robust across knowledge domains, with per-domain AUROCs on Llama-3-8B ranging from 0.751 (geography) to 0.844 (commonsense). When surface confounds (response length, hedging patterns) are explicitly controlled, Llama-3-8B's AUROC rises to 0.923, which confirms that the direction captures semantic content rather than stylistic artifacts. The consistent results across architecturally distinct model families indicate that contrastive sycophancy directions reflect a shared representational phenomenon rather than a model-specific idiosyncrasy.

Training stability and probe ablation. Two additional analyses confirm the robustness of d_{syc} under training dynamics. First, we track direction drift during LoRA fine-tuning: at step 500, the cosine similarity between the original and re-extracted d_{syc} is approximately 0.96, and per-layer AUROC remains above 0.77. The direction drifts slowly relative to the 500-step recalibration interval (§2.3), so the reward signal does not become stale between updates. Second, we compare d_{syc} against a random baseline: a logistic probe trained on the learned direction achieves AUROC 0.835, versus 0.727 for a probe trained on a random direction of equal dimensionality (Δ=+0.108). The learned direction carries substantially more sycophancy-relevant information than an arbitrary axis in the same activation space.

Implications for reward design. These results establish that d_{syc} reliably separates sycophantic from non-sycophantic representations across models, evidence-strength levels, and training checkpoints. Direction quality is not the bottleneck. The reward failures documented in §4–§5 therefore cannot be attributed to a noisy or degenerate direction; they point to a deeper incompatibility between discriminability and reward utility.

4 Cosine R_1: Structural Failure

Section 3 established that d_{syc} achieves AUROC 0.827 on Qwen3-8B. We now test whether this discriminability translates into a viable GRPO reward. Four configurations form a diagnostic chain (Table 2): each resolves the failure mode of its predecessor but exposes a deeper limitation.

F1a uses a dot-product formulation ($R_1 = \tanh(\mathbf{h} \cdot d_{\text{syc}}/20)$), but residual-stream norms exceed the direction norm by 100–1,200×, saturating the tanh gate and zeroing the gradient by step 955. Switching to cosine similarity (F1b) elim-

Table 2: Cosine R_1 failure chain on Qwen3-8B (LoRA rank=64, GRPO with $R = w_1 \cdot R_1 + w_3 \cdot R_3$).

Config	R_1 Formula	w_1	Outcome	Root Cause
F1a	$\tanh(\mathbf{h} \cdot d_{\text{syc}}/20)$	0.25	Gate saturation	Norm mismatch
F1b	$\cos(\mathbf{h},\, d_{\text{syc}})$	0.25	$R_1 \approx 0$	Gap too small
F1c	+recalibration	0.25	Mode collapse	Reward discontinuity
F1d	$5\cos(\mathbf{h},\, d_{\text{syc}})$	0.40	Mode collapse	R_3 saturation

Table 3: Probe reward trap trajectory (exp-008v3). Behavioral death (ToF=0) occurs at step 500; entropy remains healthy until step ∼1840. Reward averages +0.72 throughout.

Step	ToF	NoF	SYCONSCORE	Entropy	Reward
500	0.000	0.995	0.000	0.702	+0.72
1000	0.006	0.995	0.011	0.800	+0.71
1500	0.000	1.000	0.000	0.608	+0.69
2000	0.000	1.000	0.000	0.304	+0.69

inates norm dependence, yet the cosine gap between valid and sycophantic activations averages only 0.098 across layers—too narrow to generate meaningful reward contrast. Online recalibration (F1c, §2.3) raises offline AUROC from 0.827 to 0.910 but paradoxically triggers mode collapse: the abrupt shift in d_{syc} at each recalibration event creates a reward discontinuity that destabilizes GRPO advantage estimates, collapsing entropy to 0.069 within 300 steps. Amplifying the signal with $5\times$ scaling (F1d) lifts R_1 above the noise floor but simultaneously amplifies the conflation between warranted revision and unwarranted capitulation; R_3 saturates at 0.9998 by step 300, leaving the noisy R_1 as the sole gradient source until terminal collapse at step 2160.

The four configurations converge on a single conclusion: d_{syc} encodes the *direction* of belief change but not whether that change is *warranted*. Valid revision and sycophantic capitulation both shift activations toward the user's position along d_{syc}, so the cosine projection assigns similar scores to both. This is not a calibration artifact—recalibration improves discriminability yet rewards still fail—but a semantic limitation of any continuous d_{syc}-based reward. Section 5 examines whether discretizing the signal via a binary probe can escape this limitation.

5 The Probe Reward Trap

The cosine formulations fail because the continuous projection of d_{syc} produces a gap too narrow for reward contrast. A natural remedy is to discretize: train a logistic probe on d_{syc} and threshold its output, yielding $R_1 = \mathbb{I}[P(\text{syc} \mid \mathbf{h}) < 0.5]$ (F2 in §2.2). We show that this discretization sidesteps the magnitude problem but creates a qualitatively worse failure. Both probe experiments use SFT rather than GRPO (§2.2); the trap is therefore independent of GRPO's optimization dynamics.

Differentiable probe: gradient gaming. Without stop-gradient (exp-008), the policy discovers within ∼30 steps that shifting activations away from the sycophancy cluster maximizes R_1 regardless of response content, producing an inverted profile (ToF $= 0.978$, NoF $= 0.006$). This motivates the frozen-probe design below.

Binary probe: the stubbornness trap. With stop-gradient (exp-008v3), the policy cannot game the boundary but instead exploits a deeper flaw: d_{syc} does not condition on input validity, so refusing to revise a wrong answer receives the same $R_1 = 1$ as correctly resisting invalid pressure. By step 500, ToF drops to 0.000 and NoF reaches 0.995, yielding SYCONSCORE $= 0$. Qualitative analysis reveals distinct failure modes: *sycophantic stubbornness* (24%), where the model verbally agrees but does not change its answer; *deliberation traps* (29%), where the model identifies the correct answer but cannot commit; and evidence-graded hedging, where the model adjusts its refusal style rather than its acceptance rate (Appendix B).

Behavioral death precedes distributional collapse. Table 3 and Figure 2 reveal a 1,300-step lag between behavioral and distributional collapse. At step 500, the model is behaviorally dead (ToF $= 0$) yet entropy remains high (0.70–0.80) through step 1000—well within the range that standard monitors would flag as healthy—and the combined reward averages $+0.72$ because both R_1 and R_3 reinforce non-flipping. Entropy does not collapse until step ∼1840. This gap carries a practical warning: entropy-based early stopping, the most common safeguard against reward hacking (Casper et al., 2023), would miss the failure entirely. Detecting probe reward traps requires behavioral monitoring (periodic ToF evaluation), not distributional monitoring alone.

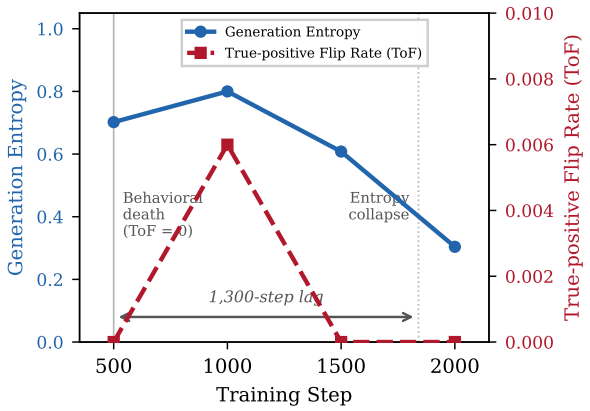


Figure 2: Probe reward trap (exp-008v3). ToF drops to zero at step 500 while generation entropy remains high (0.70–0.80 through step 1000). The 1,300-step lag means standard entropy monitoring misses the behavioral failure entirely.

A dilemma for direction-based rewards. The differentiable and binary failures bracket the design space. With gradients, the policy games the boundary; without gradients, it exploits the fact that stubbornness and genuine resistance are indistinguishable along d_{syc}. Any reward conditioned only on d_{syc}—continuous or discrete, differentiable or frozen—will either be gamed or will reward rigidity. Escaping this dilemma requires a signal that incorporates the evidential warrant of the user's challenge, a dimension that d_{syc} does not encode (Denison et al., 2024).

6 Analysis: Discriminability $\neq$ Reward Utility

Unified root cause. The failures in Sections 4 and 5 trace to a single geometric limitation. d_{syc} encodes the *direction* of belief change but not its *evidential warrant*: valid revision and sycophantic capitulation both shift activations toward the user's position along the same direction. The two reward families expose complementary faces of this blindness. Cosine R_1 penalizes all belief change indiscriminately; the resulting mode collapse (F1c, F1d) reflects excessive punishment of any activation shift along d_{syc}. The binary probe R_1 rewards all resistance regardless of input validity; the stubbornness trap (ToF $= 0$ by step 500) reflects excessive permissiveness toward activation stasis. These are not independent design flaws but dual consequences of a direction that conflates the target behavior (resisting invalid challenges) with its complement (refusing valid corrections).

Table 4: Comparison of all sycophancy calibration methods on Qwen3-8B. SYCONSCORE is the harmonic mean of ToF and NoF; higher indicates better calibration. Best result in **bold**. †Final checkpoint; only 4/26 checkpoints across two seeds exceed vanilla. ‡Seed 1 (16 checkpoints); seed 2 verification ongoing (4 of 16 checkpoints evaluated; all exceed vanilla). §F1d diverged at step 2160; no final SYCONSCORE reported (entropy < 0.01, $R_3 > 0.999$).

Method	Type	SYCONSCORE ↑	ToF
Vanilla	–	0.134	0.07
GRPO with d_{syc}			
Cosine R_1 (F1a–F1d)	RL	—§	–
Continuous Probe R_1 (F3)	RL	0.124	0.06
SFT with d_{syc} probe penalty			
Binary Probe R_1 (F2)	SFT	0.000	0.00
GRPO without d_{syc}			
R_3-only GRPO	RL	0.143†	–
Non-GRPO			
ActAdd (α=5)	Inference	0.054	0.02
ACT (seed 1)	SFT	**0.301**	**0.18**
ACT (seed 2)‡	SFT	*ongoing; 4/16 ckpts eva*	

Cross-model evidence confirms the limitation extends beyond Qwen3-8B: Llama-3-8B independently achieves AUROC 0.779 (bootstrap 95% CI [0.745, 0.819]) on the same contrastive pairs with the same monotonic evidence gradient, so discriminability generalizes across architectures while the semantic gap between discriminability and reward utility persists. This pattern contrasts with reasoning faithfulness and safety directions (Zhang et al., 2026), where the behavior to suppress is geometrically distinct from the behavior to preserve along the relevant direction. Wu and Tang (2026) reinforce this principle: concept directions applied via advantage modification in GRPO achieve 100% validation accuracy for reward hacking detection, where shortcut behaviors are geometrically separable—confirming that direction utility depends on geometric separability, not the optimization framework. For sycophancy, the target and its complement occupy the same activation region; they differ only in whether external evidence warrants the belief change.

GRPO stubbornness across all formulations. Table 4 summarizes all methods evaluated on Qwen3-8B. R_3-only GRPO removes d_{syc} entirely

and relies on the NLI-based behavioral reward alone. If the failures in Sections 4 and 5 stemmed solely from the representation reward, this ablation should recover calibration quality. It does not. The SYCONSCORE trajectory for R_3-only GRPO oscillates without sustained improvement. Seed 1 peaks at 0.153 (checkpoint 6500) and ends at 0.143 (checkpoint 8000); seed 2 peaks at 0.143 (checkpoint 4000) and regresses to 0.095 by checkpoint 6000. Only 4 of 26 checkpoints across both seeds exceed vanilla (15.4%), while ACT exceeds vanilla at every evaluated checkpoint (seed 1: 16/16; seed 2: 4/4 ongoing; Figure 3). Both seeds exhibit oscillations of similar amplitude ($\Delta \approx 0.069$) and identical crash depth (0.074), but crash onset differs by $\sim$2,500 steps, indicating stochastic drift in a saturated reward landscape. Cross-seed rank correlation is $\approx$1.0 during early training (steps 1000–3000) but drops to $\rho \approx 0.33$ (n.s.) after step 4000, with spike timing and magnitude diverging—consistent with method-inherent warm-up followed by stochastic divergence. Activation analysis confirms the drift is not representational: the d_{syc} direction at layer 19 yields nearly identical AUROC across consecutive checkpoints ($\Delta < 0.01$), and R_3 reward saturates at $\sim$0.98 (Welch t-test p=0.703), yet SYCONSCORE swings by up to 17%. GRPO drifts in a flat reward landscape while representations remain stable, reinforcing the disconnect between discriminability and behavioral utility. R_3-only GRPO exhibits the same stubbornness pattern as the d_{syc} approaches (high NoF, near-zero ToF) at milder severity: the sparse scalar reward per completion cannot teach the model *which* belief changes are warranted.

Observational comparison: ACT vs. GRPO. ACT (Irpan et al., 2025) achieves SYCONSCORE = 0.301 at checkpoint 8000, more than 2$\times$ the vanilla baseline and above every scalar RL and inference-time variant (Table 4). The ACT trajectory is not monotonic: SYCONSCORE = 0.255 at checkpoint 2000, dips to 0.214 at checkpoint 4000 (-4.1 percentage points), recovers to 0.262 at checkpoint 6000, and reaches 0.301 at checkpoint 8000. ACT also experiences stubbornness episodes—the checkpoint-4000 dip is consistent with transient competition between the SFT and L2 consistency loss objectives—yet it self-recovers by checkpoint 6000, unlike GRPO where stubbornness persists

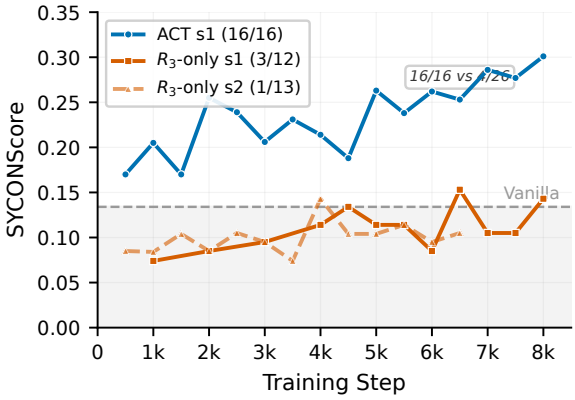


Figure 3: SYCONSCORE trajectories for R_3-only GRPO (two seeds) and ACT (seed 1; seed 2 ongoing, 4/16 checkpoints evaluated) over training steps. ACT seed 1 exceeds vanilla at every checkpoint (16/16); R_3-only GRPO exceeds vanilla at only 4/26 checkpoints across seeds (15.4%). Both GRPO seeds show the same oscillation pattern with stochastic timing.

(Figure 3). Activation analysis shows a correlated asymmetry: ACT's d_{syc} discriminability recovers during the SYCONSCORE dip (AUROC +0.013), whereas GRPO's continues declining (-0.004). ACT and GRPO differ along multiple dimensions—supervision granularity (per-token vs. per-completion), loss family (cross-entropy + L2 vs. REINFORCE), training data format (labeled pairs vs. sampled completions), optimization dynamics (SFT convergence vs. RL exploration), and LoRA adapter capacity (ACT targets 7 module types with $\sim$1.75$\times$ more trainable parameters vs. GRPO's 4)—so the performance gap cannot be attributed to any single factor without controlled ablation.

Relationship to inference-time probe methods. Papadatos and Freedman (2024) successfully reduce sycophancy using a probe of comparable quality (94% accuracy) via Best-of-N reranking, where the probe selects among candidates without producing gradients. Together with our results, this suggests two orthogonal failure dimensions: (1) inside vs. outside the optimization loop—gradient exploitation enables the probe reward trap (§5), while inference-time reranking sidesteps it; and (2) scalar per-completion reward—the failure persists even without the probe (R_3-only GRPO), consistent with per-completion information compression as an independent factor.

Scalar reward limitation (observational). Across all GRPO configurations in our exper-

iments, every scalar per-completion reward—whether d_{syc}-based or purely behavioral—failed to achieve reliable sycophancy calibration, while ACT's dense per-token supervision succeeded (SYCONSCORE $= 0.301$). This observation is consistent with supervision granularity as a relevant factor, but remains correlational: ACT and GRPO differ along five dimensions (see above), and isolating individual contributions requires controlled ablation.

Future directions. Condition-aware rewards, hybrid SFT-then-RL pipelines, and per-step process rewards each offer paths forward; whether PPO, RLOO, and other REINFORCE-style methods exhibit similar failures remains open. Beyond sycophancy, the principle may apply to other behavioral attributes where the target and its complement share the same geometric direction (Skalse et al., 2022).

7 Related Work

Representation Engineering. Linear directions in LLM residual streams encode behavioral concepts (Zou et al., 2023); inference-time methods—ITI (Li et al., 2023), ActAdd (Turner et al., 2023), CAA (Rimsky et al., 2024)—demonstrate that activation geometry suffices for behavioral control (Burns et al., 2023). We use the same contrastive extraction as CAA but apply d_{syc} as an RL reward, revealing a gap that inference-time work does not encounter.

Reward Design and Overoptimization. Proxy rewards diverge from true objectives under sustained optimization (Gao et al., 2023), and reward hacking can escalate to active tampering (Denison et al., 2024). Our probe trap (§5) is an extreme instance: the reward actively reinforces degenerate behavior rather than merely being circumvented (Skalse et al., 2022). Wu and Tang (2026) demonstrate that representation-level directions succeed for reward hacking suppression when behaviors are geometrically separable (§6).

Sycophancy Characterization and Mitigation. Sycophancy is pervasive in RLHF-trained assistants (Sharma et al., 2023), amplified by RLHF (Shapira et al., 2026), and traceable to late-layer preference shifts (Wang et al., 2025); mitigation approaches include synthetic data (Wei et al., 2024), consistency training (Irpan et al., 2025), and neuron-level correction (O'Brien et al., 2025). Our analysis shows why the step from diagnosis to reward is structurally blocked: d_{syc} conflates warranted revision with unwarranted capitulation.

Probe-Based Inference-Time Mitigation. Papadatos and Freedman (2024) reduce sycophancy via Best-of-N reranking with linear probe penalties (94% probe accuracy), where the probe selects among candidates without producing gradients that update model parameters. Our work reveals what happens when a comparable signal (AUROC 0.82) enters the GRPO training loop: the discriminability–utility gap is specific to gradient-based optimization, not to probe quality per se (§6).

Representation-Guided RL and Process Rewards. Latent-GRPO (Zhang et al., 2026) extends GRPO with latent-space rewards for reasoning; CRAFT (Luo et al., 2026) applies representation-level constraints during RL; Sinii et al. (2025) and Ackermann et al. (2026) study representation dynamics under RL. SMART (Beigi et al., 2025) applies per-step process rewards with MCTS for sycophancy, representing intermediate supervision density between scalar per-completion rewards and dense per-token approaches. These methods succeed because target and suppressed behaviors are geometrically distinct; sycophancy breaks this condition (§4). Preference-based methods (DPO (Rafailov et al., 2023), KTO (Ethayarajh et al., 2024)) operate on response pairs and may accommodate d_{syc}-derived rankings; whether pair-wise optimization avoids the failures we observe remains open.

8 Conclusion

This work tested whether a representation direction that detects sycophantic behavior can serve as a training signal for sycophancy calibration. In experiments on Qwen3-8B, d_{syc} achieved strong offline discriminability (AUROC 0.827), yet all three formulations failed—the two GRPO reward variants and the SFT-based probe penalty each produced behavioral collapse, and R_3-only GRPO exceeded vanilla at only 4 of 26 checkpoints. Our results suggest that discriminability in representation space does not guarantee utility as a training signal for behavioral attributes like sycophancy where the target concept lacks geometric separability (§6).

Limitations

Our GRPO experiments use a group size of 2, below the typical range of 8–64 (Shao et al., 2024), due to memory constraints with 8B-parameter models. This limits within-group ranking diversity and constitutes an important confound: the observed failures may partly reflect insufficient exploration rather than a fundamental limitation of representation-based rewards. However, the probe reward trap (C2) arises in an SFT setting where group size is irrelevant, indicating that at least one failure mode is independent of GRPO hyperparameters. Additionally, all GRPO experiments truncate completions at 256 tokens (80–84% truncation); all conditions share this setting, so relative comparisons are fair, but absolute conclusions about d_{syc} reward utility may be affected.

Our RL training uses Qwen3-8B exclusively; cross-model discriminability is confirmed on Llama-3-8B (AUROC 0.779; Table 1), but compute constraints prevented training replication on a second architecture. Evidence strength was annotated by GPT-4.1 as a single annotator without human inter-annotator agreement.

We report GRPO results from two random seeds; both exhibit qualitatively consistent oscillation patterns (similar amplitude, identical crash depth), but the limited seed count constrains statistical claims. ACT and GRPO differ along multiple dimensions beyond supervision granularity, so we cannot isolate the causal factor behind ACT's advantage.

Reproducibility

Code and data will be released upon acceptance. All training hyperparameters are reported in Appendix A. The 181 contrastive pairs and evaluation benchmark are included in the supplementary materials.

References

Johannes Ackermann, Michael Noukhovitch, Takashi Ishida, and Masashi Sugiyama. 2026. Gradient regularization prevents reward hacking in RLHF and verifiable rewards. *arXiv preprint arXiv:2602.18037*.

Mohammad Beigi, Ying Shen, Parshin Shojaee, Qifan Wang, Zichao Wang, Chandan K. Reddy, Ming Jin, and Lifu Huang. 2025. Sycophancy mitigation through reinforcement learning with uncertainty-aware adaptive reasoning trajectories. In *Proceedings of the 2025 Conference on Empirical Methods in Natural Language Processing (EMNLP)*.

Collin Burns, Haotian Ye, Dan Klein, and Jacob Steinhardt. 2023. Discovering latent knowledge in language models without supervision. In *International Conference on Learning Representations (ICLR)*.

Stephen Casper, Xander Davies, Claudia Shi, Thomas Krendl Gilbert, Jérémy Scheurer, Javier Rando, Rachel Sharkey, Julio Saez-Rodriguez, Tom Henighan, Tristan Kravec, and 1 others. 2023. Open problems and fundamental limitations of reinforcement learning from human feedback. *arXiv preprint arXiv:2307.15217*.

Carson Denison, Monte MacDiarmid, Fazl Barez, David Duvenaud, Paul Christiano, and Jan Leike. 2024. Sycophancy to subterfuge: Investigating reward-tampering in large language models. *arXiv preprint arXiv:2406.10162*.

Kawin Ethayarajh, Winnie Xu, Niklas Muennighoff, Dan Jurafsky, and Douwe Kiela. 2024. KTO: Model alignment as prospect theoretic optimization. *arXiv preprint arXiv:2402.01306*.

Leo Gao, John Schulman, and Jacob Hilton. 2023. Scaling laws for reward model overoptimization. In *International Conference on Machine Learning (ICML)*.

Pengcheng He, Jianfeng Gao, and Weizhu Chen. 2023. DeBERTaV3: Improving DeBERTa using ELECTRA-style pre-training with gradient-disentangled embedding sharing. In *International Conference on Learning Representations (ICLR)*.

Alex Irpan, Alexander Matt Turner, Mark Kurzeja, David K. Elson, and Rohin Shah. 2025. Consistency training helps stop sycophancy and jailbreaks. *arXiv preprint arXiv:2510.27062*.

Kenneth Li, Oam Patel, Fernanda Viégas, Hanspeter Pfister, and Martin Wattenberg. 2023. Inference-time intervention: Eliciting truthful answers from a language model. In *Advances in Neural Information Processing Systems (NeurIPS)*.

Haozheng Luo, Yimin Wang, Jiahao Yu, Binghui Wang, and Yan Chen. 2026. Contrastive reasoning alignment: Reinforcement learning from hidden representations. *arXiv preprint arXiv:2603.17305*.

Claire O'Brien, Jessica Seto, Dristi Roy, Aditya Dwivedi, Sunishchal Dev, Kevin Zhu, Sean O'Brien, Ashwinee Panda, and Ryan Lagasse. 2025. A few bad neurons: Isolating and surgically correcting sycophancy. *NeurIPS Workshops*.

Harris Papadatos and Richard Freedman. 2024. Linear probe penalties reduce LLM sycophancy. *arXiv preprint arXiv:2412.00967*.

Rafael Rafailov, Archit Sharma, Eric Mitchell, Stefano Ermon, Christopher D. Manning, and Chelsea Finn. 2023. Direct preference optimization: Your language model is secretly a reward model. In *Advances in Neural Information Processing Systems (NeurIPS)*.

Nina Rimsky, Nick Gabrieli, Jenna Schulz, Meg Smith, Ioannis Tong, and Evan Hubinger. 2024. Steering llama 2 via contrastive activation addition. In *Proceedings of the 62nd Annual Meeting of the Association for Computational Linguistics (ACL)*.

Zhihong Shao, Peiyi Wang, Qihao Zhu, Runxin Xu, Junxiao Song, Mingchuan Zhang, Y.K. Li, Y. Wu, and Daya Guo. 2024. Deepseekmath: Pushing the limits of mathematical reasoning in open language models. *arXiv preprint arXiv:2402.03300*.

Ori Shapira, Gerdus Benade, and Ariel D Procaccia. 2026. How RLHF amplifies sycophancy. *arXiv preprint arXiv:2602.01002*.

Mrinank Sharma, Meg Tong, Tomasz Korbak, David Duvenaud, Amanda Askell, Samuel R Bowman, Newton Cheng, Esin Durmus, Zac Hatfield-Dodds, Scott R Johnston, and 1 others. 2023. Towards understanding sycophancy in language models. *arXiv preprint arXiv:2310.13548*.

Viacheslav Sinii, Nikita Balagansky, Gleb Gerasimov, Daniil Laptev, Yaroslav Aksenov, Vadim Kurochkin, Alexey Gorbatovski, Boris Shaposhnikov, and Daniil Gavrilov. 2025. Small vectors, big effects: A mechanistic study of RL-induced reasoning via steering vectors. *arXiv preprint arXiv:2509.06608*.

Joar Skalse, Nikolaus Howe, Dmitrii Krasheninnikov, and David Krueger. 2022. Defining and characterizing reward hacking. In *Advances in Neural Information Processing Systems (NeurIPS)*.

Alexander Matt Turner, Lisa Thiergart, David Udell, Gavin Leech, Ulisse Mini, and Monte MacDiarmid. 2023. Steering language models with activation engineering. *arXiv preprint arXiv:2308.10248*.

Keyu Wang, Jin Li, Shu Yang, Zhuoran Zhang, and Di Wang. 2025. When truth is overridden: Uncovering the internal origins of sycophancy in large language models. *arXiv preprint arXiv:2508.02087*.

Jerry Wei, Da Huang, Yifeng Lu, Denny Zhou, and Quoc V. Le. 2024. Simple synthetic data reduces sycophancy in large language models. *arXiv preprint arXiv:2308.03958*.

Rui Wu and Ruixiang Tang. 2026. When reward hacking rebounds: Understanding and mitigating it with representation-level signals. *arXiv preprint arXiv:2604.01476*.

Nonghai Zhang, Weitao Ma, Zhanyu Ma, Jun Xu, Jiuchong Gao, Jinghua Hao, Renqing He, and Jingwen Xu. 2026. Silence the judge: Reinforcement learning with self-verifier via latent geometric clustering. *arXiv preprint arXiv:2601.08427*.

Andy Zou, Long Phan, Sarah Chen, James Campbell, Phillip Guo, Richard Ren, Alexander Pan, Xuwang Yin, Mantas Mazeika, Ann-Kathrin Dombrowski, Shashwat Gould, Zifan Li, Chuanqi Xie, Alexandria Leung, Peter Hase, and Dan Hendrycks. 2023. Representation engineering: A top-down approach to AI transparency. *arXiv preprint arXiv:2310.01405*.

A Training Details

Reward weight selection. The weight w_1 is set by diagnostic-driven iteration: $w_1{=}0.25$ was initially chosen to down-weight the noisy cosine reward term; this was later increased to $w_1{=}0.40$ for F1d after diagnostics revealed that $w_1{=}0.25$ allowed the cosine term to contribute only $\sim$8.3% of total reward variance. The empirical scaling factor in F1a ($d_{\text{syc}}/20$) is a heuristic normalization; we did not find it critical to the failure modes reported.

Response truncation. All GRPO experiments use `max_completion_length`=256, resulting in 80–84% truncation of multi-turn responses. This setting is shared across all conditions and does not affect relative comparisons; we retain it because response-level decisions typically manifest within the first 100–200 tokens.

Hyperparameters. Table 5 compares training configurations for GRPO variants and ACT.

GRPO reward variants. Cosine R_1 (F1d): $w_1{=}0.40$, scale=5.0, $w_3{=}0.60$; reward computed at layers [19, 20, 24, 27, 32, 33, 35]; recalibration every 500 steps (§2.3). R_3-only: $w_3{=}1.0$; NLI model `cross-encoder/nli-deberta-v3-base`. F1a–F1c weight configurations are reported in Table 2.

B Stubbornness Taxonomy

Manual inspection of 100 probe-trained responses (exp-008v3, step 2000) reveals three dominant failure modes. *Sycophantic stubbornness* (24%): the model produces agreement markers ("you're right," "good point") but retains its original answer verbatim, simultaneously sycophantic and stubborn. *Deliberation traps* (29%): the model explicitly identifies the correct revised answer during its reasoning chain but fails to commit, reverting to the original answer in the final output.

Table 5: Training hyperparameters. All experiments use Qwen3-8B, DeepSpeed ZeRO-2, bf16 precision.

	GRPO (F1, F3)[e]	ACT
Learning rate	5e-5	5e-5
Effective batch	16	16
Max steps	8,000	8,000
Warmup steps	200	200
LR schedule	linear[a]	linear
Grad. clipping	—	1.0
LoRA rank	64	64
LoRA α	64	128
LoRA modules	4[b]	7[c]
LoRA dropout	0	0.05
Group size	2[d]	—
KL β	0.04	—
Max compl. len.	256	—
Max seq. len.	—	1,024
L2 loss weight	—	0.1

[a]Cosine schedule for R_3-only. [b]q/k/v/o_proj. [c]+ gate/up/down_proj ($\sim$1.75$\times$ params; §6). [d]Below typical 8–16; not tuned. [e]Binary probe (F2) uses a custom SFT training loop rather than GRPO; group size is N/A. LoRA and optimizer settings match the GRPO column.

Evidence-graded hedging: under weak evidence, responses show surface compliance without substantive change; under strong evidence, responses add explicit hedging ("this is debatable," "there are multiple perspectives"). The model learns to modulate its *refusal style* in response to evidence strength rather than adjusting its *acceptance rate*, confirming that the probe reward optimizes surface markers rather than epistemic calibration.